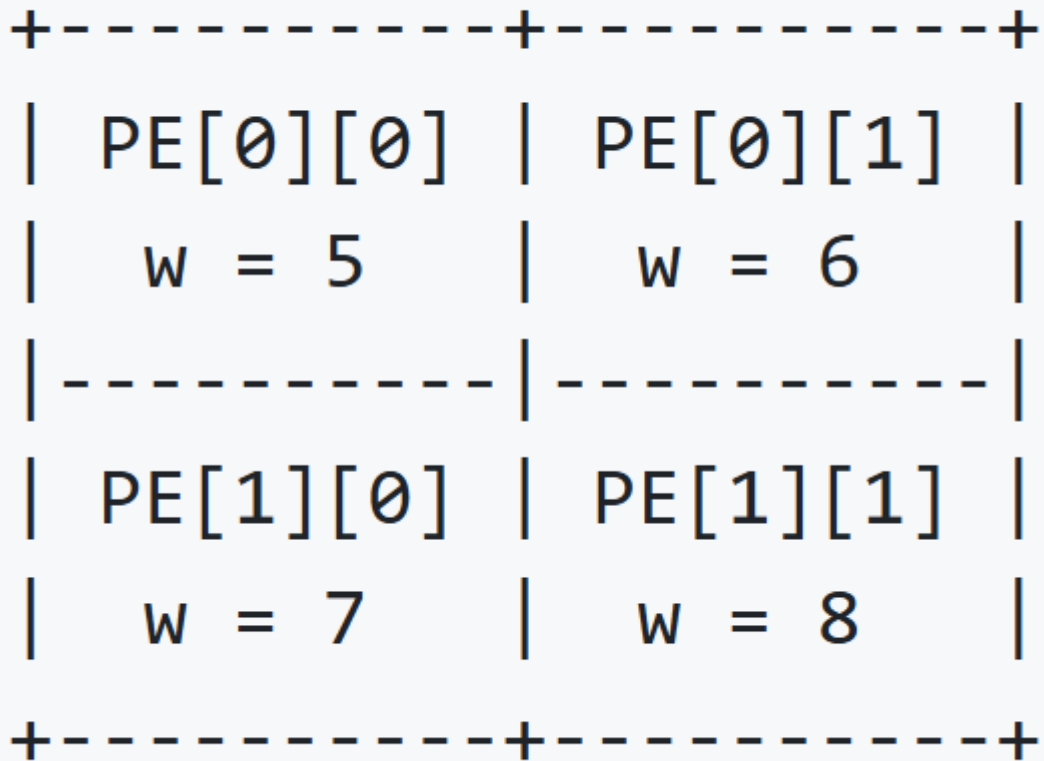


CODEFEST - 5

TASK 1: Systolic Array PE Diagram



TASK 2. Cycle-by-Cycle Trace Table

The following table traces the progression of partial sums (PS) to compute $C = A \times B$.

- **Matrix A:** $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$
- **Matrix B:** $\begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$

	A	B	C	D	E	F	G	H
1	Cycle	Input Row 0	Input Row 1	PE[0][0] (PS)	PE[0][1] (PS)	PE[1][0] (PS)	PE[1][1] (PS)	Output C
2	t=0	a11=1	-	1*5 = 5	-	-	-	-
3	t=1	a12=2	a21=3	3*5 = 15	1*6 = 6	5 + (2*7) = 19	-	-
4	t=2	-	a22=4	-	3*6 = 18	15 + (4*7) = 43	6 + (2*8) = 22	C11=19
5	t=3	-	-	-	-	-	18 + (4*8) = 50	C21=43, C12=22
6	t=4	-	-	-	-	-	-	C22=50

Final Result C: $\begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$

TASK 3. Computation Metrics

- **(a) Total MAC operations performed: 8**
- **(b) Input value reuse:** Each input value is reused 2 times.
- **(c) Number of off-chip memory accesses:**
 - **Matrix A (Reads): 4**
 - **Matrix B (Preload): 4**
 - **Matrix C (Writes): 4**
 - **Total Accesses: 12**

TASK 4. Output-Stationary Comparison

In an output-stationary dataflow, the partial sums (the values for Matrix C) stay fixed inside the PEs and accumulate there while both A and B values stream in from memory every cycle.